Inference, Homework 1, and Simple Linear Regression

ISyE 4031, Week 2 Class 3

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Today: inference, Homework 1, setup, and regression

0–15 min	Finish p-values, rejection regions, errors, and assumptions.
15–30 min	Walk through Homework 1: ownership, delegation, testing, and submission.
30–50 min	Install Cursor, sign in, open a project, and verify that Agent works.
50–70 min	Introduce the simple linear regression model and its interpretation.
70–75 min	Confirm forecasting teams and complete the exit check.

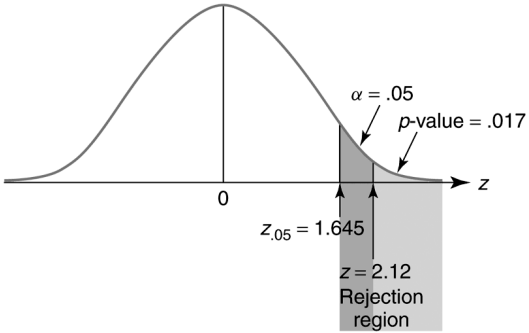
Statistical work includes a reproducible workflow.

A p-value measures compatibility with the null model

Assuming H_0, the p-value is the probability of a statistic at least as extreme as the observed statistic.

It is **not**:

- $P(H_0 \text{ is true} \mid \text{data})$;
- the probability the result occurred by chance;
- the size or importance of an effect.



A p-value does not make the decision for us

Suppose a two-sided test gives

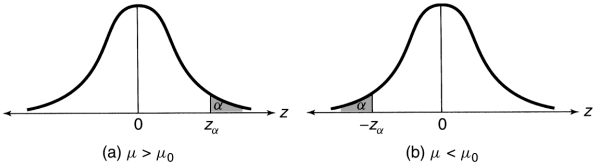
$$t_{\text{obs}} = 2.31, \qquad df = 18, \qquad p = 0.033.$$

$\alpha = 0.05$
Reject H_0 because $0.033 < 0.05$.

$\alpha = 0.01$
Do not reject H_0 because $0.033 > 0.01$.

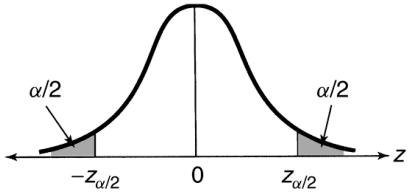
Report the effect, uncertainty, assumptions, and decision threshold.

The alternative determines the rejection region



(a) $\mu > \mu_0$ (b) $\mu < \mu_0$

One-sided alternatives



(c) $\mu \neq \mu_0$

Two-sided alternative

Choose the direction before looking at the data.

Type I error, Type II error, and power encode tradeoffs

	H_0 true	H_a true
Do not reject H_0	correct	Type II, β
Reject H_0	Type I, α	power, $1-\beta$

- Type I error probability: α.
- Type II error probability: β.
- Power: $1-\beta$.

Power depends on effect size, noise, sample size, the test, and α.

Assumptions are part of the answer

Independence	Observations and errors do not carry unmodeled dependence.
Distribution	The selected normal, t, or other approximation is adequate.
Stable process	The mechanism does not shift between the sample and target.
Legal information	No future values enter a forecast or its features.

When an assumption fails, change the method or narrow the claim.

Homework 1 asks you to build, verify, and explain

Parts 1–3

Calculate, visualize, forecast, and interpret with the legal information set.

Part 4

Diagnose the flawed agent analysis, replace it, and prove key properties with tests.

Submission

Deliver a reproducible ZIP, not disconnected answers or unexplained output.

Correct numbers are necessary; defensible reasoning and reproducibility are graded too.

Delegate production tasks, not statistical responsibility

The agent may

- propose code and explanations;
- suggest tests and edge cases;
- help debug failed commands;
- review a draft for inconsistencies.

You remain responsible for

- target, unit, and information cutoff;
- assumptions and calculations;
- interpretations and conclusions;
- every accepted or rejected suggestion.

If you cannot explain a result without the agent, it is not ready to submit.

Your ZIP should run in a clean course environment

Five files at the top level

- `analysis.py` or a notebook
- `test_analysis.py`
- `report.pdf`, at most five pages
- `AI_USAGE.md`
- `README.md`

Run before uploading

```
python analysis.py
pytest -q
```

Submit only `hw1_lastname_firstname.zip`.

Another person should reproduce every reported result without editing your code.

Part 4 turns agent output into auditable evidence

1. **Inspect:** identify at least five distinct statistical, forecasting, or reproducibility problems.
2. **Explain:** state the consequence of each problem, not only the faulty line.
3. **Repair:** replace the analysis while keeping the transparent baselines separate.
4. **Test:** demonstrate cutoff safety, reproducibility, baseline behavior, and required identities.
5. **Document:** record delegation, tests, and one accepted, revised, or rejected suggestion in `AI_USAGE.md`.

The audit trail is evidence of judgment, not a transcript dump.

Install Cursor and verify one local-agent workflow

1. Download the desktop application from `cursor.com/download`.
2. Run the installer, open Cursor, and sign in.
3. Open a small course folder or a local copy of the Homework 1 files.
4. Open Agent and ask it to **inspect first and propose a plan**.
5. Review the proposed files and commands before approving changes.

Account check

Cursor Hobby has limited Agent requests. If your account displays a university or event promotion, verify its terms in your billing dashboard before relying on it.

Cursor installation and plan details: cursor.com/download and cursor.com/pricing, checked August 31, 2026.

Hands-on checkpoint: make one safe request

Before we move on, confirm that you can:

- launch Cursor and open a course folder;
- open Agent and see the selected model or mode;
- distinguish a proposed change from an applied change;
- find the terminal and review a command before it runs.

First prompt

```
Read the assignment and the files in this folder.
Do not change anything yet. Summarize the required
deliverables, then propose a step-by-step work plan.
```

Regression turns association into a conditional model

We observe an outcome Y and a predictor X:

$$Y = f(X) + \varepsilon.$$

- $f(X)$ is the systematic signal we try to learn.
- ε contains variation the model does not explain.
- Prediction combines the estimated signal with honest uncertainty.

Nowcasting example

Use timely surveillance signals X to estimate current influenza activity Y before the official value is complete.

One predictor gives simple linear regression

For observations $i = 1, \ldots, n$,

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

- Y_i response or dependent variable;
- x_i predictor, explanatory variable, or feature;
- β_0 intercept: mean response when $x = 0$;
- β_1 slope: change in mean response for a one-unit increase in x;
- ε_i unobserved error around the conditional mean.

Linear means linear in the coefficients

Linear models

$$\beta_0 + \beta_1 x$$

$$\beta_0 + \beta_1 x^2$$

$$\beta_0 + \beta_1 \log x$$

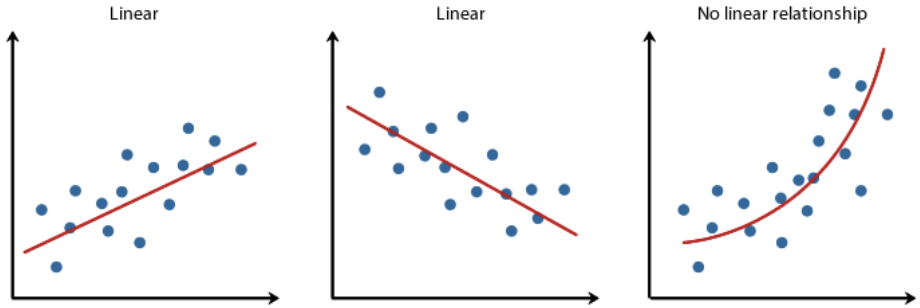
Nonlinear in parameters

$$\beta_0 + e^{\beta_1 x}$$

$$\frac{\beta_0 x}{\beta_1 + x}$$

Transforming the predictor can bend the fitted curve while preserving linearity in the coefficients.

Different patterns require different models



- The first two patterns can be summarized by a straight conditional mean.
- The third pattern has strong association, but a straight line misses its shape.

The fitted line estimates a conditional mean

Population relationship:

$$E(Y_i \mid X_i = x_i) = \beta_0 + \beta_1 x_i.$$

Estimated relationship:

$$\widehat{Y}_i = b_0 + b_1 x_i.$$

Interpretation boundary

The slope describes association in the modeled population. It is not automatically a causal effect, and it may not remain valid outside the observed range.

Form a forecasting team this week

Team structure

- Target: 12 teams of 3 to 4 students.
- Build complementary strengths.
- Rotate data, modeling, operations, and evaluation roles.

Before the next class

- Identify preferred teammates.
- If you need matching, say so early.
- Be ready to work in one reproducible repository.

A team is a forecasting operation, not a division of disconnected tasks.

Exit check: specify and interpret the model

For $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$:

1. What does β_1 describe, and which units are required?
2. What is random in the model, and what is treated as observed?
3. Why is association not automatically causation?
4. What pattern in a scatterplot would challenge a straight conditional mean?

Next: least squares, residuals, and a complete fitted line.

Optional: the five-point example exposes every residual

For $\widehat{y} = -0.1 + 0.7x$:

x	y	$\widehat{y}$	$e = y - \widehat{y}$	e^2
1	1	0.6	0.4	0.16
2	1	1.3	−0.3	0.09
3	2	2.0	0.0	0.00
4	2	2.7	−0.7	0.49
5	4	3.4	0.6	0.36
		$\sum$	0.0	1.10

Optional: a small p-value can accompany a small effect

Large data set

A tiny effect can be estimated precisely and produce a small p-value.

Small data set

A practically important effect can remain uncertain and produce a large p-value.

Report together:

- effect estimate and units;
- confidence interval;
- p-value and test definition;
- assumptions and practical consequence.

Appendix Box 1: Inspect and plan

```
The current project contains one file: fuel_consumption.csv.
Do not create or change files yet.

1. Inspect the CSV. Report its columns, number of rows, data types,
   missing values, and observed ranges.
2. Identify the temperature predictor and fuel-consumption response.
3. State the regression estimand and the assumptions needed to
   interpret a simple linear regression.
4. Propose a short, reproducible implementation plan.
5. List numerical and conceptual checks that should be completed
   before the analysis is accepted.

Wait for my approval before creating or editing files.
Do not use the internet or add external data.
```

Appendix Box 2: Implement the regression

```
Proceed with the approved plan.

1. Create analysis.py and parse fuel_consumption.csv.
2. Do not silently remove, replace, or reorder observations.
3. Fit ordinary least squares with temperature as the predictor,
   fuel consumption as the response, and an intercept.
4. Independently calculate:
      b1 = Sxy / Sxx
      b0 = y_bar - b1 * x_bar
5. Verify that the hand calculation and fitted model agree.
6. Calculate fitted values, residuals, SSE, and MSE.
7. Calculate the point prediction at 70 degrees Fahrenheit and
   determine whether it is interpolation or extrapolation.

Run analysis.py and show the numerical results.
Do not use the internet, external data, or an unexplained formula.
```

Appendix Box 3: Audit and test

```
Audit the implementation as a skeptical statistical reviewer.

1. Create test_analysis.py.
2. Test agreement between the formula-based and fitted coefficients.
3. Test that residuals sum to approximately zero.
4. Test that the fitted line passes through (x_bar, y_bar).
5. Test that the mean fitted value equals y_bar.
6. Check that all CSV rows were handled explicitly and that missing
   values, if any, were not silently discarded.
7. Run analysis.py and all tests. Show the commands and results.
8. If a check fails, diagnose the cause before changing the code.

Passing these tests does not prove that a straight-line model is
scientifically adequate. State what the tests do and do not verify.
```

Appendix Box 4: Interpret and report

```
Complete the analysis and produce an evidence-based walkthrough.

1. Create a labeled scatterplot with the fitted regression line.
2. Report the fitted equation with units.
3. Interpret the slope as a change in conditional mean response.
4. Explain whether the intercept has a useful interpretation.
5. Explain the limitation of the prediction at 70 degrees Fahrenheit.
6. Explain why the slope shows association and does not, by itself,
   establish causality.
7. List the created files, numerical results, test evidence,
   assumptions, and unresolved limitations.

Do not claim that successful execution or passing tests proves the
model is correct. Distinguish computation, verification, and judgment.
```